

Lab Report for CSE 4412: Data Communication & Networking

Lab 08: DHCP, Static Routing, RIP, OSPF, and Core Router Configuration

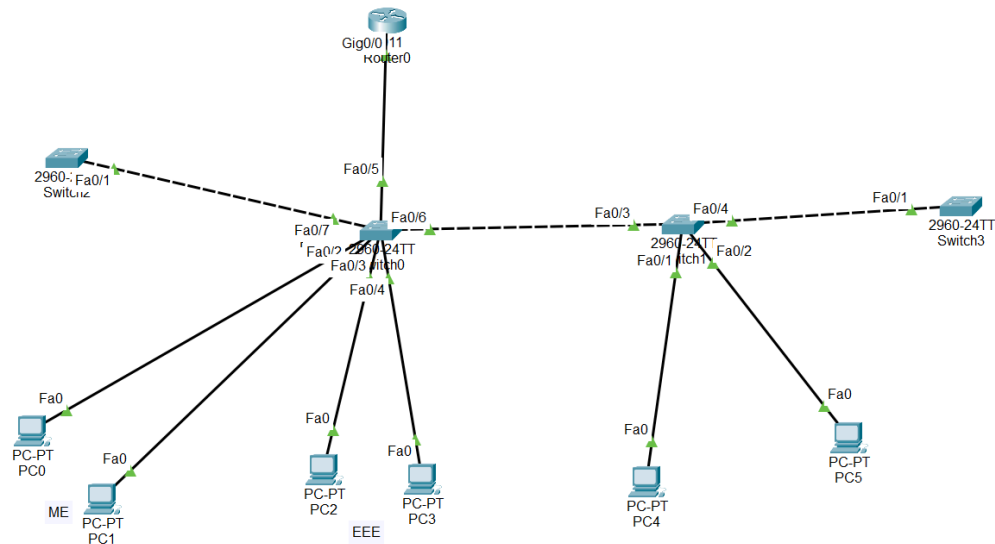
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1. Lab Task(s)

- Implement a network scenario with three labs:
 - **Lab Alpha:** 2 PCs, static routing, subnet `192.168.id.0/24`.
 - **Lab Beta:** 16 PCs, RIP routing, subnet `192.168.(id+1).0/24`.
 - **Lab Charlie:** 16 PCs, OSPF routing, subnet `192.168.(id+2).0/24`.
- Connect all labs to a **Core Router** which re-routes information to the ICT department.
- Configure **DHCP** so that all devices receive IP addresses automatically.
- Verify connectivity between PCs across the labs and with the ICT department.

2. Screenshot of the Final Network Topology



3. Procedure

1. Network Design

- Drew three separate LANs:
 - **Lab Alpha:** One router + 2 PCs.
 - **Lab Beta:** One router + switch + 8 PCs.
 - **Lab Charlie:** One router + switch + 8 PCs.
- Connected each lab router to the **Core Router**

2. IP Addressing & DHCP

- Configured **DHCP pools** on each lab router:
 - Lab Alpha → **192.168.id.0/24**
 - Lab Beta → **192.168.(id+1).0/24**
 - Lab Charlie → **192.168.(id+2).0/24**

- Verified that all PCs automatically received IP addresses via DHCP.

3. Routing Configuration

- **Lab Alpha Router:** Configured **static routing** towards the Core Router.
- **Lab Beta Router:** Configured **RIP v2** for dynamic routing. Advertised its subnet to Core Router.
- **Lab Charlie Router:** Configured **OSPF** with a unique process ID. Advertised its subnet to Core Router.
- **Core Router:** Configured with redistribution to handle routing updates from static, RIP, and OSPF so that all labs could communicate with each other.

4. Testing

- Used **ping** and **tracert** commands from PCs in different labs to ensure end-to-end connectivity.
- Verified that IP addresses were auto-assigned from DHCP.
- Confirmed that routing tables on the Core Router contained entries for all three lab subnets.

4. Observations & Results

- All PCs successfully obtained IP addresses automatically via DHCP.
- **Lab Alpha PCs** could reach Lab Beta and Lab Charlie PCs through the Core Router.
- RIP and OSPF correctly propagated routes to the Core Router.
- Core Router acted as a central point for routing updates and enabled full connectivity to ICT Dept.
- Packet Tracer simulation mode confirmed correct routing protocols in action.

5. Challenges Faced

- Initial issue with **DHCP pools overlapping**; resolved by carefully defining non-overlapping address ranges.
- Configuring **route redistribution** on the Core Router was tricky since it needed to understand static, RIP, and OSPF together. Fixed by explicitly redistributing between protocols.
- OSPF areas needed to be consistent; once configured properly, routing stabilized.